Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/11/2005
PI Name:	Gratzl, Miklos Ph.D.		
Project Title:	Microminiature Monitor for Vital Electrolyte and Metabolite Levels of Astronauts		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106-7207	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	08/01/2002	**End Date:**	07/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2807		
Performance Goal No.:			
Performance Goal Text:			

Ions, such as proton (pH) and potassium, play a crucial role in body fluids to maintain proper basic functioning of cells and tissues. Metabolites, such as glucose, control the energy available to the entire human body in normal as well as stress situations, and before, during, and after meals. These molecules diffuse easily between blood in the capillaries and the interstitial fluid residing between cells and tissues. The monitor proposed in this grant represents a synthesis of unique concepts to achieve accurate, continuous in vivo monitoring of critical ions (called electrolytes) and glucose in the interstitial fluid under the human skin. Proton and potassium levels will be monitored using optode technology that translates the respective ionic concentrations into variable colours of corresponding ionophore/dye/polymeric liquid membranes. Glucose will be monitored indirectly, by coupling through immobilized glucose oxidase with pH that is then detected using a similar colour scheme. The monitor will consist of a tiny plastic bar, 100-200 µm wide and 1-2 mm long, placed just under the skin, with optical spots or stripes for each analyte as well as blanks. The colours will be read and translated into concentration values by a watch-like device placed above the skin. Direct optical coupling between

Task Description:	the in vivo sensing bar and the ex vivo detector device requires negligible power, and eliminates the need for wires or optical fibres crossing the skin. The microminiature bar penetrates the skin easily and painlessly, so that astronauts could insert it themselves. The approach is fully compatible with telemetry in space, and thus, in vivo clinical data will be available real time in the Earth based command center. The information provided by the device can be used for collecting hitherto unavailable vital data on clinical effects of space travel. Managing clinical emergencies in space with the sensor already in place should also become much more efficient than without a continuous monitor, as is currently the case. Civilian applications may include better glucose control of patients with moderate to sever diabetes: a growing health problem in the US and Worldwide. Monitoring of vital body fluid chemistry of astronauts during space flight requires regular collection, and in vitro analysis, of blood samples. It is desirable to replace this multistep, and invasive, procedure with an automatic and continuous in vivo diagnostic monitoring scheme. To date, no such approach is available even in terrestrial settings. We have conceived, and tested in our laboratory, an entirely new set of ideas that has performed beyond expectation in preliminary laboratory tests and that makes it possible to realize such a scheme for the first time, both for terrestrial and space applications. The approach is painless, does not require the collection of blood samples, needs negligible energy, and does not require water, reagents, or labor. It is fully compatible with telemetry between the miniature multisensing element placed under the skin, and the on-board computer system. No danger of track infection exists. In this research, we will aim at optimizing the already proven set of concepts for continuous monitoring of glucose, and basic electrolytes (pH and K+ in this project) in the interstitial fluid under the skin. The expected outcome of this research will be a fully functional, microminiature, optical multisensing bar to be placed under the skin, and a small detector unit on the skin, which will allow the continuous monitoring of glucose, pH and K+ in astronauts during space flights. An individual bar will be functional for at least three consecutive days, after which a new bar can be inserted by the astronaut under his/her skin with a simple, painless, and safe procedure. Spin-offs for other NASA projects are already being explored. Civilian applications are expected for diabetes management, and other clinical purposes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	• By developing a small scale spray dry system, microscopic optode beads as well as enzyme immobilized beads 2 - 5 micrometer in diameter have been fabricated reproducibly. • Biocompatible and physically robust microminiature poly(HEMA-co-PEGMA)-based sliver bodies have been developed with photopolymerization and excimer laser drilling techniques. These techniques and the spray dry method for micro-beads fabrication will allow the reproducible mass production of the microminiature optical monitor. • With adding an optical white reference as one of the capsules of the microminiature optical monitor, eventual instabilities in illumination light intensity and other optical interferences have been eliminated. • Tuning optode chemistry to achieve desired sensitivity for in vivo monitoring of electrolytes has been achieved by changing the amount of ionophore in the respective microscopic optode beads. • Tuning mass transport in the glucose sensing capsule for broad dynamic range has been achieved. • By adding polyethylene glycol to the stuffing mixture and the sensing wells, response time for both glucose and electrolytes has greatly improved. • pH sensitive near infrared dyes were successfully synthesized, which is expected to improve the performance of color recognition of sliver sensors placed deeper under the skin. • Mathematical tools and software has been developed for automatic evaluation of colors monitored by a CCD camera and the associated three color dimensions that we call 3D color shape analysis. • Instrumentation for in vivo optical data analysis has been designed and tested. A prototype of a small ex vivo detecting device combined with color CCD, macro lens and illuminator has been developed. • Good biocompatibility and encouraging in vivo functionality have been obtained by using poly(HEMA) as sensor body material. • In vivo functionality tests in rats are currently being evaluated with the 3D color shape approach.
Bibliography Type:	Description: (Last Updated: 10/21/2009)
Articles in Peer-reviewed Journals	Tohda K, Gratzl M. "Micro-miniature autonomous optical sensor array for monitoring ions and metabolites 2: color responses to pH, K+ and glucose." Anal Sci. 2006 Jul;22(7):937-41. PubMed PMID: 16837742 , Jul-2006
Articles in Peer-reviewed Journals	Tohda K, Gratzl M. "Micro-miniature autonomous optical sensor array for monitoring ions and metabolites 1: design, fabrication, and data analysis." Anal Sci. 2006 Mar;22(3):383-8. PubMed PMID: 16733308 , Mar-2006